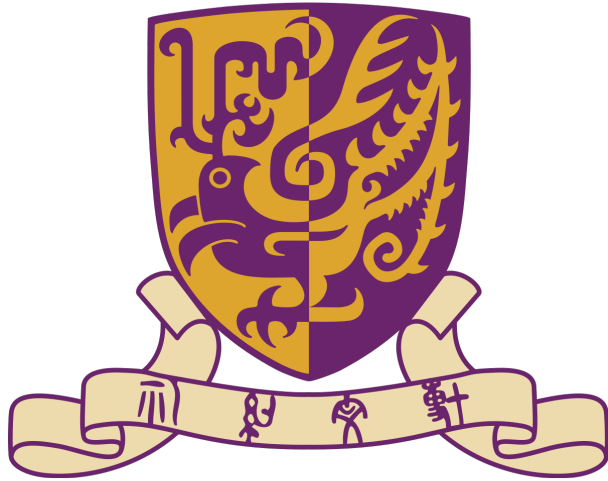




THE CHINESE UNIVERSITY OF HONG KONG,
SHENZHEN

ENG 2001

English for Academic Purposes II

Self-Confidence in Math and STEM Major Choice at CUHK-Shenzhen

PAN, Jeremiah

November 28, 2025

Contents

1	Introduction	2
2	Research Questions	3
3	Methodology	3
3.1	Participants	3
3.2	Materials	3
3.3	Procedures	4
3.4	Data Analysis	4
4	Results	5
4.1	Descriptive Statistics of Key Variables	5
4.2	RQ1 — Is there a relationship between students' math self-confidence and their choice of a STEM major?	6
4.3	RQ2 — Does math self-confidence predict actual math performance?	6
4.4	RQ3 — Are Chinese students more confident in math than international students?	7
4.5	RQ4 — What external factors affect students' math confidence?	8
5	Discussion	9
5.1	Summary of Key Findings	9
5.2	Interpretation of Findings	9
5.3	Limitations	10
5.4	Suggestions for Future Research	10
6	Conclusion	11

Abstract

Math self-confidence plays a critical role in students' STEM major selection — yet its relationship with performance and cultural background remains understudied in Sino-foreign universities, where flexible major-switching policies are common. While existing literature suggests that East Asian students often report lower confidence despite high achievement, no prior study has examined whether this pattern persists among Chinese and international undergraduates within an internationally oriented Chinese university. To address this gap, I surveyed 91 CUHK-Shenzhen undergraduates using a quantitative instrument measuring math confidence, self-reported performance, and influencing factors. Results indicated that STEM majors reported significantly higher math confidence than non-STEM majors, and confidence moderately predicted actual math performance; however, no significant difference emerged between Chinese and international students. External influences were dominated by academic experiences — particularly past exam results and peer comparison — rather than cultural background. These findings challenge assumptions about cultural determinants of confidence and highlight the importance of academic environment in shaping STEM choices, offering educators actionable insights to support students' academic decision-making.

1 Introduction

Choosing a STEM major is shaped not only by academic performance but also by students' confidence in their mathematical abilities (Escobar et al., 2023; Wang, 2013). While prior research confirms that math self-confidence predicts STEM pursuit (Moakler Jr and Kim, 2014) and that academic confidence often diverges from actual performance (Ferdinand et al., 2024), few studies examine how this dynamic plays out in Sino-foreign cooperative universities in mainland China — particularly in institutions like The Chinese University of Hong Kong, Shenzhen (CUHK-Shenzhen), where students have greater freedom to choose or change majors.

Existing literature also highlights cultural differences in math confidence: East Asian students often excel academically but report lower self-assurance due to high expectations and norms of humility (Leung, 2017). However, most comparative studies focus on national education systems or Western contexts. None explicitly compares Chinese and international students within an internationally oriented university in China — such as CUHK-Shenzhen, which hosts approximately 14–15% international students (Shaw College Master, personal communication, November 2025). Nor do they account for structural features like delayed major declaration (for Chinese students at CUHK-Shenzhen), which may shape confidence over time.

This study fills that gap. It investigates how math self-confidence influences STEM major choice at CUHK-Shenzhen, comparing Chinese and international undergraduate students. It also explores whether confidence correlates with performance and what external factors affect confidence levels. By focusing on the CUHK-Shenzhen context, this research provides actionable insights for educators seeking to support diverse student populations in making confident, informed decisions — beyond grades alone.

2 Research Questions

This study investigates how math self-confidence influences STEM major choice at CUHK-Shenzhen, with a focus on differences between Chinese and international students. Guided by the research gap identified above, the following questions are addressed (All questions are examined specifically within the CUHK-Shenzhen context):

1. Is there a relationship between students' math self-confidence and their choice of a STEM major?
2. Does math self-confidence predict actual math performance?
3. Are Chinese students more confident in math than international students?
4. What external factors (e.g., peer influence, parental expectations) affect students' math confidence?

3 Methodology

This study employed a quantitative survey design to examine the research questions. Data were collected via an online questionnaire distributed through Microsoft Forms between October 22 and November 14, 2025.

3.1 Participants

A convenience sample of nearly 100 undergraduate students enrolled at CUHK-Shenzhen was recruited, including both Chinese and international students from STEM and non-STEM majors. Participants were primarily recruited through two channels: (1) sharing the survey QR code via WeChat Moments and student group chats (targeting Chinese students), and (2) in-person distribution by international student peers, who shared the link with their friends via communication apps (targeting international students).

3.2 Materials

The survey consisted of four main sections:

- **Demographics:** Multiple-choice questions capturing nationality, year of study, and declared major.
- **Math Performance:** Self-reported high school math grades, Gaokao math score, and first-year university math course grades. To facilitate correlation analysis, these were combined into a weighted composite score (60% university grade, 40% high school grades), based on the assumption that recent performance better reflects current ability (Sax et al., 2015). For first-year students who had not yet completed a university math course, the high school math score was weighted at 100%.
- **Math Confidence:** A 9-item scale measuring confidence across calculation, proof-solving, real-world application, and self-perception, adapted from Moakler Jr and Kim (2014). Each item was rated on a 5-point Likert scale (1 = Strongly Disagree to 5 = Strongly Agree).

- **Influencing Factors:** Multiple-choice items (select all that apply) assessing seven perceived influences on math confidence. Additionally, one item measured the self-reported importance of math confidence in major selection, rated on a 5-point scale (Not at all important to Extremely important).

All items were designed specifically for this study and are included in full in the Appendix.

The variables of interest and their corresponding survey questions are summarized in Table 1.

Table 1: Survey Variable Categories and Measurement Types

Variable Category	Question(s)	Measurement Type
Demographics	Q2–Q4	Categorical
Math Performance	Q6–Q8	Categorical
Math Self-Confidence	Q5	Ordinal (Likert)
Cultural Factors	Q9	Ordinal (Likert)
Influencing Factors	Q10	Categorical (Multiple Selection)
Importance of Confidence	Q11	Ordinal
STEM Choice (consideration + rationale)	Q12–Q13	Categorical
Open-Ended Feedback	Q14	Qualitative

3.3 Procedures

The survey was administered online using Microsoft Forms. Participants accessed the questionnaire via a QR code or direct link. The survey took approximately 5–7 minutes to complete.

3.4 Data Analysis

Quantitative data were analyzed using SPSS (Version 28). Descriptive statistics were computed for all variables. Inferential analyses included:

- **Independent samples t-tests** to compare math confidence scores between: (a) Chinese and international students, and (b) STEM and non-STEM students.
- **Pearson correlation coefficient** to measure the linear relationship between the composite math performance score and math confidence scores.
- **Descriptive statistics** for influencing factors, presented as frequency counts and percentages, visualized using bar charts.

All statistical tests were conducted at a significance level of $p < 0.05$.

Participation in this study was voluntary, anonymous, and confidential. All participants provided informed consent prior to completing the survey. Data were collected and stored securely via Microsoft Forms.

4 Results

4.1 Descriptive Statistics of Key Variables

The sample consisted of 91 undergraduate students at CUHK-Shenzhen. Demographically, 75 (82.4%) were Chinese students and 16 (17.6%) were international students. In terms of major, 56 (61.5%) were STEM majors and 35 (38.5%) were Non-STEM majors. By year of study, 20 (22.0%) were Year 1, 26 (28.6%) were Year 2, 30 (33.0%) were Year 3, 13 (14.3%) were Year 4, and 2 (2.2%) were in “Other” categories (e.g., those taking gap years or experiencing graduation delays).

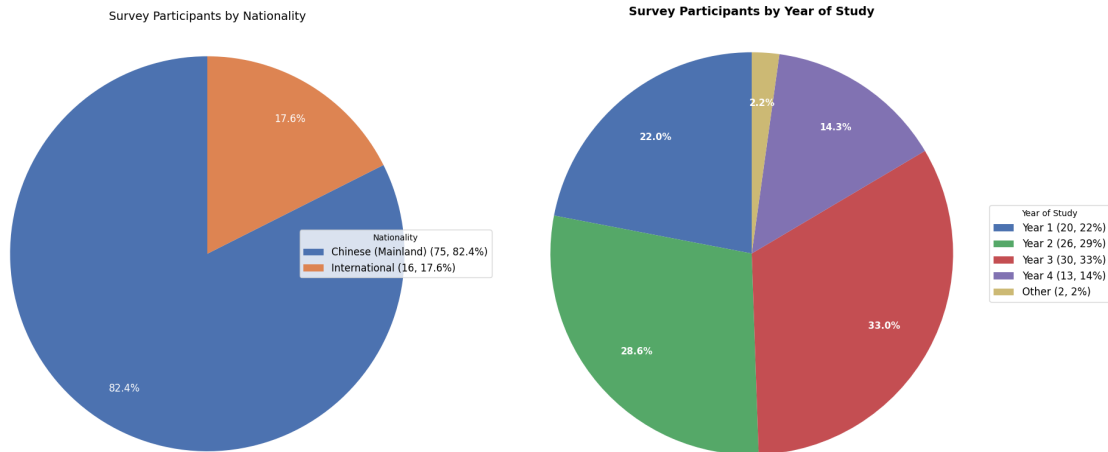


Figure 1: Demographic distribution by nationality. Figure 2: Distribution of respondents by year of study.

Table 2: Demographic composition of the sample ($N = 91$).

Category	Count	Percentage
Nationality		
Chinese	75	82.4%
International	16	17.6%
Major		
STEM	56	61.5%
Non-STEM	35	38.5%
Year of Study		
Year 1	20	22.0%
Year 2	26	28.6%
Year 3	30	33.0%
Year 4	13	14.3%
Other	2	2.2%

The overall mean math self-confidence score was 3.07 ($SD = 0.82$), with scores ranging from 1.11 to 4.78. Math performance scores had a mean of 3.22 ($SD = 0.85$), ranging from 0.50 to 4.50. The composite cultural factor score (average of 4 items) had a mean of 3.68 ($SD = 0.75$).

Table 3: Descriptive statistics for key variables ($N = 91$).

Variable	Mean	SD	Range
Math Confidence	3.07	0.82	1.11–4.78
Math Performance	3.22	0.85	0.50–4.50
Cultural Factor Score	3.68	0.75	1.78–5.00

4.2 RQ1 — Is there a relationship between students' math self-confidence and their choice of a STEM major?

To address this question, I compared math self-confidence scores between STEM and Non-STEM majors. As shown in Table 4, STEM students ($n = 56$) reported significantly higher math confidence ($M = 3.26$, $SD = 0.83$) than Non-STEM students ($n = 35$, $M = 2.76$, $SD = 0.72$), $p = 0.003$, significant ($p < 0.05$). This suggests that students who choose STEM majors tend to have higher math self-confidence. Figure 3 visually confirms this difference through box plots.

Table 4: Math confidence by major group ($N = 91$).

Group	Mean (SD)	N	p -value
STEM	3.26 (0.83)	56	0.003
Non-STEM	2.76 (0.72)	35	

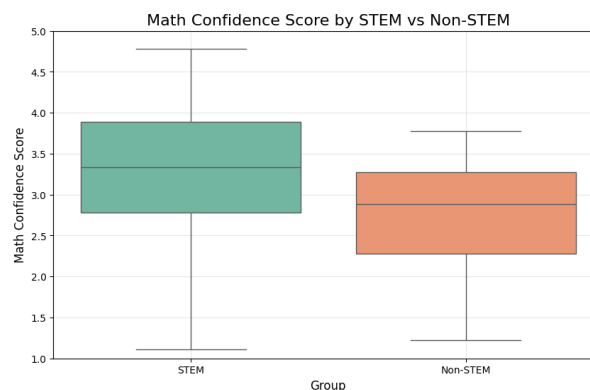


Figure 3: Math confidence scores by major group.

4.3 RQ2 — Does math self-confidence predict actual math performance?

A Pearson correlation analysis was conducted to examine the relationship between math self-confidence and math performance. As illustrated in Figure 4, there is a statistically significant, moderate positive correlation between the two variables ($r = 0.455$, $p < 0.001$, $N = 91$). This indicates that higher math self-confidence is associated with better math performance.

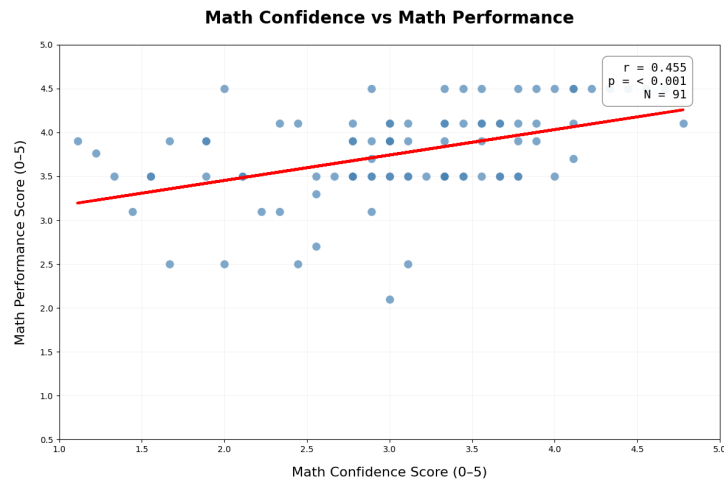


Figure 4: Scatter plot of math confidence vs. math performance ($r = 0.455$, $p < 0.001$).

4.4 RQ3 — Are Chinese students more confident in math than international students?

To investigate whether nationality influences math self-confidence, I compared Chinese ($n = 75$) and International students ($n = 16$). As shown in Table 5, no significant difference was found between the two groups ($M_{Chinese} = 3.08$, $SD = 0.85$; $M_{International} = 3.01$, $SD = 0.67$), $p = 0.705$, not significant ($p > 0.05$). Figure 5 further illustrates the similarity in distributions, suggesting that nationality does not significantly affect math confidence in this sample, though the relatively small sample of international students ($n = 16$) warrants cautious interpretation.

Table 5: Math confidence by nationality ($N = 91$).

Group	Mean (SD)	N	p -value
Chinese	3.08 (0.85)	75	0.705
International	3.01 (0.67)	16	

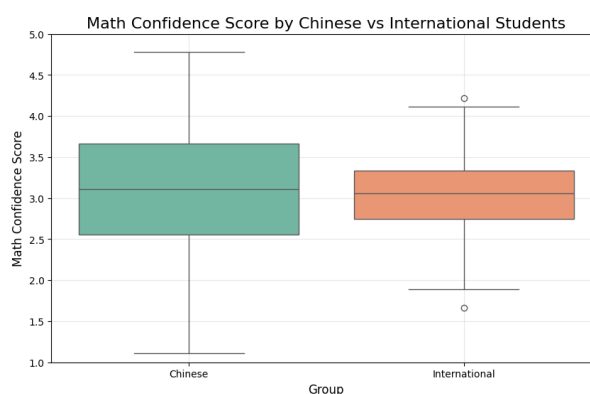


Figure 5: Math confidence scores by nationality.

4.5 RQ4 — What external factors affect students' math confidence?

I examined the frequency of selected influencing factors using a multi-select question. As presented in Table 6 and visualized in Figure 6, the most frequently cited factor was *Past exam results* (75.8%), followed by *Peer comparison* (62.6%) and *University math courses* (41.8%). In contrast, *Cultural background* was selected by only 7.7% of respondents.

Table 6: Frequency and percentage of influencing factors on math confidence ($N = 91$).

Factor	Count	Percentage
Past exam results	69	75.8%
Peer comparison	57	62.6%
University math courses	38	41.8%
Personal interest in math	40	44.0%
Teacher feedback	26	28.6%
Parental expectations	19	20.9%
Cultural background	7	7.7%
Other	0	0.0%

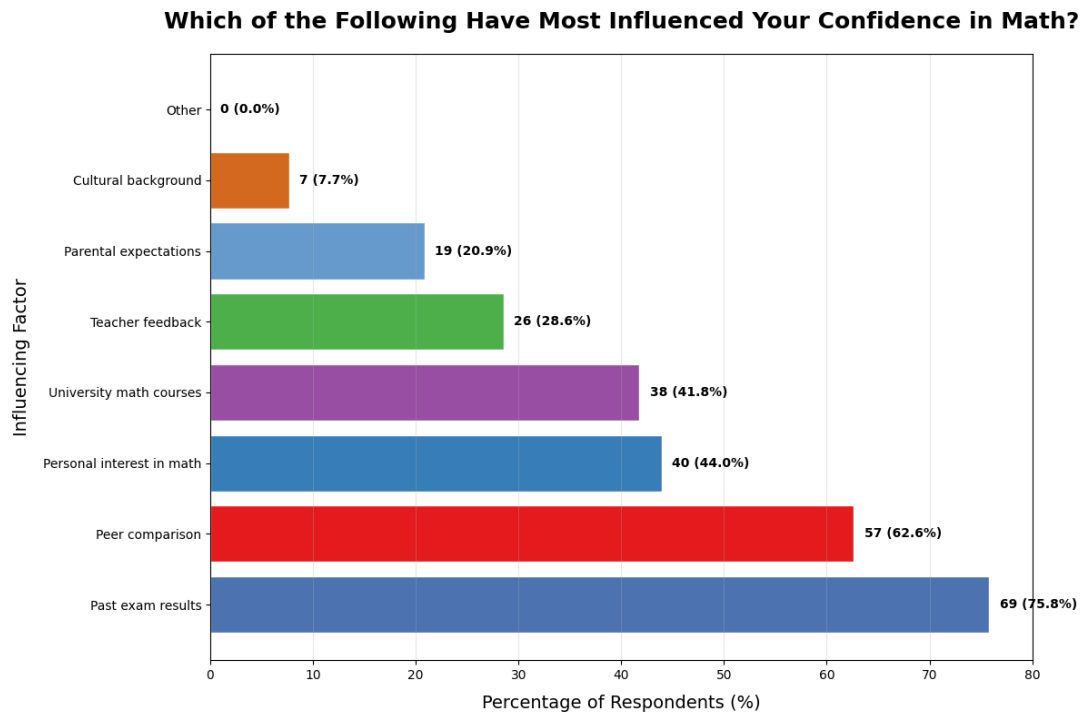


Figure 6: Influencing factors on math confidence (multiple responses allowed).

5 Discussion

5.1 Summary of Key Findings

The results indicate that:

- Math self-confidence is significantly higher among STEM majors (RQ1).
- Math confidence moderately predicts math performance (RQ2).
- No significant difference exists between Chinese and international students in math confidence (RQ3).
- Academic experiences (past exams, peer comparisons) are the primary influencing factors of math confidence, not cultural factors (RQ4).

5.2 Interpretation of Findings

The finding that STEM majors reported significantly higher math self-confidence than Non-STEM majors ($p = 0.003$) aligns with prior research suggesting that self-efficacy and perceived ability are key predictors of STEM persistence (Moakler Jr and Kim, 2014; Escobar et al., 2023). This is particularly meaningful in the context of CUHK-Shenzhen, where students enjoy greater freedom to choose or change majors. The absence of bureaucratic barriers means that students' choices more accurately reflect their self-perceived ability — reinforcing the idea that confidence, not just grades, drives major selection. In a university setting that emphasizes student autonomy, confidence becomes a critical filter for major selection — a finding that has practical implications for academic advising and career counseling.

The moderate positive correlation between math self-confidence and math performance ($r = 0.455, p < 0.001$) supports the idea that confidence is a useful way to predict academic success. While some research suggests that confidence does not always match actual ability—especially when students are overconfident or anxious (Ferdinand et al., 2024)—my results show a clearer link in this specific context. This supports studies that find a two-way relationship: confidence can improve grades because students try harder, while good grades make students feel more confident. However, because the correlation is moderate rather than strong, it shows that "feeling good at math" and "getting high grades" are still two different things, confirming that confidence is not exactly the same as ability.

One of the most intriguing findings of this study is the absence of a statistically significant difference in math confidence between Chinese and international students ($p = 0.705$). This contradicts the widely held belief — supported by cross-national studies — that East Asian students, despite high academic achievement, report lower self-assurance due to cultural norms of humility and high expectations (Leung, 2017). However, this result may be explained by the unique context of CUHK-Shenzhen. As an internationally oriented university in mainland China, it creates a micro-environment where cultural boundaries blur. International students may internalize the perception that "Chinese students are good at math," leading them to benchmark themselves against a high standard — thereby reducing the gap in self-reported confidence. This interpretation is supported by the influencing factors data: 62.6% of students selected "Peer

comparison” as a key influence on their math confidence, while only 7.7% selected “Cultural background.” This suggests that in the CUHK-Shenzhen context, immediate academic environment and social comparisons outweigh broader cultural norms in shaping confidence—a phenomenon consistent with the “Big-Fish-Little-Pond Effect,” where high-ability environments can paradoxically lower academic self-concept (Marsh et al., 2008).

The overwhelming dominance of academic experiences — particularly “Past exam results” (75.8%) and “Peer comparison” (62.6%) further reinforces the idea that confidence is shaped more by recent, tangible academic experiences than by abstract cultural or familial expectations. This is consistent with Bandura’s theory of self-efficacy, which identifies “mastery experiences” and “social comparison” as the two most powerful sources of self-efficacy beliefs (Bandura, 2010).

5.3 Limitations

Several limitations should be considered when interpreting the findings of this study. First, the reliance on a convenience sample ($N = 91$) may restrict the generalizability of the results to the broader student population at CUHK-Shenzhen. As participation was voluntary, the sample may suffer from self-selection bias, where students with stronger interests in mathematics were more likely to respond.

Second, the way “STEM major” was defined in this study may oversimplify the reality of students’ academic paths. As shown in Table 2, the analysis divided students into two main groups: STEM and Non-STEM. However, because I grouped “declared STEM majors” together with “students considering a transfer,” and classified “undecided” students as Non-STEM, the results might miss some important details. For instance, students who are still exploring their options might have very different confidence levels compared to those committed to a Non-STEM major, which could make the comparison less accurate.

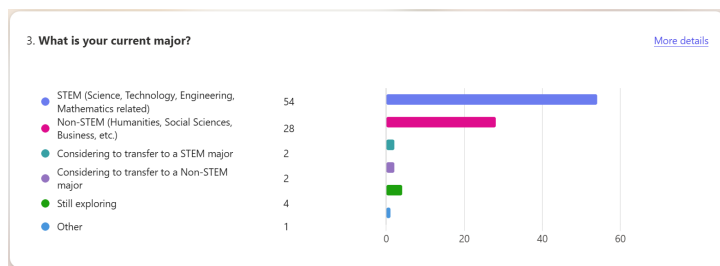


Figure 7: The original survey question on major selection.

Finally, the measurement of math performance via self-reported grades (High School, Gaokao, and University) introduces potential measurement error. Unlike administrative data, self-reported scores are subject to recall bias and social desirability bias, where participants may unintentionally or intentionally misreport their grades. Without verification against official academic transcripts, the internal validity of the performance metrics remains a constraint common to survey-based research.

5.4 Suggestions for Future Research

Future research could address these limitations by adopting a more robust methodological framework to enhance the validity and depth of the findings. Specifically, refining

the classification of academic majors to distinguish between “declared STEM majors,” “STEM intenders,” and “undecided” students—stratified by academic year—would offer a more nuanced understanding of how math confidence evolves during critical decision-making stages. This is particularly pertinent within the flexible academic environment of CUHK-Shenzhen, where major switching is common. Furthermore, the reliance on self-reported data could be mitigated by collaborating with the university registry to incorporate objective academic transcripts, thereby ensuring higher reliability in performance metrics. Complementing this with a longitudinal design would allow researchers to track the co-evolution of math confidence and performance over time, elucidating whether shifts in confidence serve as a predictor for major changes or academic outcomes.

Beyond internal design improvements, the scope of the study should be broadened to enhance generalizability. Expanding the demographic diversity of the sample, particularly by recruiting a larger cohort of international students, is essential to better characterize variations across distinct subpopulations within the university. Additionally, replicating this study across other Sino-foreign joint ventures, such as NYU Shanghai or Duke Kunshan University, would facilitate cross-institutional comparisons. Such comparative analysis is necessary to determine whether the current findings are unique to the specific pedagogical culture and policies of CUHK-Shenzhen or if they reflect broader trends inherent to the Sino-foreign educational model.

6 Conclusion

This study suggests that at CUHK-Shenzhen, math self-confidence is higher among STEM majors, moderately predicts performance, and is shaped by academic experiences — not culture. Surprisingly, no confidence gap exists between Chinese and international students, suggesting that the academic environment outweighs cultural norms.

Its significance lies in highlighting confidence as a key driver of major choice in flexible academic environments — offering actionable insights for educators to support student decision-making beyond grades.

Limitations include sample size, self-reported data, and simplified major classification. Future research should explore how confidence evolves over time across diverse institutions — especially Sino-foreign universities where cultural boundaries blur.

References

- Bandura, A. (2010). Self-efficacy. *The Corsini encyclopedia of psychology*, pages 1–3.
- Escobar, M., Majewski, H. M., Qazi, M., and Rawajfih, Y. (2023). Self-efficacy in stem. In *International Encyclopedia of Education (Fourth Edition)*, pages 388–394. Elsevier.
- Ferdinand, R., Malanchini, M., and Rimfeld, K. (2024). Mathematics interest, self-efficacy, and anxiety predict stem career choice in emerging adulthood. *npj Science of Learning*, 9(1):66.
- Leung, F. K. (2017). Making sense of mathematics achievement in east asia: Does culture really matter? In *Proceedings of the 13th International Congress on Mathematical Education: ICME-13*, pages 201–218. Springer.
- Marsh, H. W., Seaton, M., Trautwein, U., Lüdtke, O., Hau, K.-T., O’Mara, A. J., and Craven, R. G. (2008). The big-fish–little-pond-effect stands up to critical scrutiny: Implications for theory, methodology, and future research. *Educational psychology review*, 20(3):319–350.
- Moakler Jr, M. W. and Kim, M. M. (2014). College major choice in stem: Revisiting confidence and demographic factors. *The Career development quarterly*, 62(2):128–142.
- Sax, L. J., Kanny, M. A., Riggers-Piehl, T. A., Whang, H., and Paulson, L. N. (2015). The changing salience of mathematical self-concept in shaping women’s and men’s stem aspirations. *Research in Higher Education*, 56(8):813–842.
- Wang, X. (2013). Why students choose stem majors: Motivation, high school learning, and postsecondary context of support. *American educational research journal*, 50(5):1081–1121.

Appendix: Survey Questionnaire

Below is a copy of the questionnaire used in this study.

2025/11/20 16:02

Survey on Math Self-Confidence and STEM Major Choice at CUHK-Shenzhen

Survey on Math Self-Confidence and STEM Major Choice at CUHK-Shenzhen

Estimated Time to Complete: 5–7 minutes

Confidentiality Statement:

Your responses are anonymous and will be used solely for academic research purposes. No personal identifiers will be collected.

* Required

1

Please read the following before proceeding:

This survey is part of a research project for ENG2001 at CUHK-Shenzhen. The purpose is to understand how students' confidence in math influences their choice of major, particularly in STEM fields. Your participation is voluntary, and you may skip any question or stop at any time. All data will be kept confidential and used only for research analysis. By clicking "Agree," you consent to participate.

*

☐ I agree to participate

2

What is your nationality? *

☐ Chinese (Mainland)

☐ International (e.g., Hong Kong, Macau, Taiwan students; Indonesia, Russia, Germany, Greece etc.)

☐ Other

3

What is your current major? *

Y1 Undeclared students please choose your intended major unless you have NO strong preference ! (Then choose "Still Exploring")

☐ STEM (Science, Technology, Engineering, Mathematics related)

☐ Non-STEM (Humanities, Social Sciences, Business, etc.)

☐ Considering to transfer to a STEM major

☐ Considering to transfer to a Non-STEM major

☐ Still exploring

☐ Other

4

What year are you currently in? *

- ☐ Year 1
- ☐ Year 2
- ☐ Year 3
- ☐ Year 4
- ☐ Other

5

MATH SELF-CONFIDENCE *

Please indicate how much you agree or disagree with each statement about your math abilities.

	Strongly Disagree	Disagree	Neutral	Agree	Strongly agree
I am confident in my ability to solve calculation problems.	☐	☐	☐	☐	☐
I feel comfortable tackling proofs.	☐	☐	☐	☐	☐
I believe I can learn new math concepts quickly.	☐	☐	☐	☐	☐
I am confident in my ability to solve real world problems involving math.	☐	☐	☐	☐	☐
I feel anxious when I have to do math in front of others.	☐	☐	☐	☐	☐
I believe I am good at math compared to my peers.	☐	☐	☐	☐	☐
I enjoy doing math problems, even when they are challenging.	☐	☐	☐	☐	☐
I think I could succeed in a difficult math course if I tried.	☐	☐	☐	☐	☐
I often doubt my math abilities, even when I get good grades.	☐	☐	☐	☐	☐

2025/11/20 16:02

Survey on Math Self-Confidence and STEM Major Choice at CUHK-Shenzhen

6

(For International Students mainly) What was your final math grade in high school?

- ☐ A range (90–100%)
- ☐ B range (80–89%)
- ☐ C range (70–79%)
- ☐ D range (60–69%)
- ☐ F (below 60%)
- ☐ Not applicable / Don't remember

7

(Optional, for Chinese students only) What is your Gaokao math score?

- ☐ 130–150
- ☐ 110–129
- ☐ 90–109
- ☐ Below 90
- ☐ Not applicable

8

What is the BEST grade you received in your first-year university math course? **(e.g., MAT1001/1011, MAT1005, MAT2040/2042, etc, including STA2001/2001H/2003)*

- ☐ A range
- ☐ B range
- ☐ C range
- ☐ D range
- ☐ F
- ☐ Not taken yet / Not applicable

9

Cultural Factors *

How much do you agree with the following statements? Scale: Strongly Disagree to Strongly Agree)

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
My high school math teachers encouraged me to believe in my math abilities.	☐	☐	☐	☐	☐
I compare my math ability to my peers, and this affects my confidence.	☐	☐	☐	☐	☐
My parents expect me to do well in math.	☐	☐	☐	☐	☐
I feel pressure to perform well in math because of cultural expectations.	☐	☐	☐	☐	☐

10

Which of the following have most influenced your confidence in math? *

(Select all that apply)

☐ Peer comparison

☐ Cultural background

☐ Personal interest in math

☐ Past exam results

☐ University math courses

☐ Parental expectations

☐ Teacher feedback

☐ Other

2025/11/20 16:02

Survey on Math Self-Confidence and STEM Major Choice at CUHK-Shenzhen

11

How important was your confidence in math when choosing your major? *

- ☐ Moderately important
- ☐ Not at all important
- ☐ Slightly important
- ☐ Extremely important
- ☐ Very important

12

Did you consider a STEM major but ultimately choose a non-STEM major? *

- ☐ No
- ☐ Yes
- ☐ Not sure / Undecided

13

If yes, what was the main reason?*(Only if Q12 = Yes)*

- ☐ Lack of interest in STEM
- ☐ Parental/family pressure
- ☐ Career prospects
- ☐ Lack of confidence in math
- ☐ Other

2025/11/20 16:02

Survey on Math Self-Confidence and STEM Major Choice at CUHK-Shenzhen

14

(Optional) Is there anything else you'd like to share about your experience with math or your major choice?

If you'd like to share your story in more detail (e.g., why you chose/didn't choose STEM, your math journey), feel free to message me on WeChat. All conversations will remain confidential.

This content is neither created nor endorsed by Microsoft. The data you submit will be sent to the form owner.

 Microsoft Forms